

SparseStack: sharper constants, whole-sketch Nyström bounds, and adaptive frame selection

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Abstract

SparseStack can use fewer rows and fewer nonzeros per column without changing its distribution or its operator proof. We sharpen an existing all-order moment certificate from the universal constant pair $(26.5, 179)$ to $(25, 154)$, with two alternative sparsity–row tradeoffs. The improvement retains the relation between the dimension ratio and the bucket count; all 63 interval checks are certified in exact rational arithmetic. We also prove a deterministic Nyström bound $\|R\|_F^2 \leq \text{tr}(AZAZ)$ and compute its exact SparseStack expectation, yielding $\mathbb{E}\|R\|_F^2 \leq 2(\text{tr } A)^2/m$. A finite-support obstruction explains why this expected bound cannot be uniformly relative to a spectral tail. Finally, we specialize convex residual drift and diagonal estimation to independent CountSketch blocks, and use classical adaptive-versus-volume sampling to justify sketch reuse with a spectral selection price at most $k!$.

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1 Introduction

SparseStack is the sparse sketching matrix

$$\Pi = s^{-1/2} \begin{bmatrix} H_1 \\ \vdots \\ H_s \end{bmatrix} \in \mathbb{R}^{m \times n}, \quad m = sb,$$

built from s independent CountSketch blocks $H_g \in \mathbb{R}^{b \times n}$ [2, 16, 17]. Every column has exactly s nonzero entries $\pm s^{-1/2}$. The direct-model theorem of *Graded Multiplication Operators for Random-Matrix Moments* [9, Theorem 10.6 and Appendix B] proves the optimal oblivious subspace embedding (OSE) guarantee $m = O((d + \log(1/\delta))/\varepsilon^2)$, $s = O(\log(d/\delta)/\varepsilon)$ with explicit universal constants: $s < 26.5 q/\varepsilon$ and $m < 179(d + q)/\varepsilon^2$, where $q = \max\{1, \lceil \log_4(d/\delta) \rceil\}$. Its proof bounds the moment $\mathbb{E} \operatorname{tr} E^{2q}$ of the Gram error $E = U^\top \Pi^\top \Pi U - I_d$ by the vacuum entry of the $2q$ -th power of a scalar tridiagonal majorant T_q , and then bounds that entry by a single concave row-sum envelope.

This companion does three things with that machinery. First, it sharpens the final scalar step: the envelope used in [9] replaces a dimension ratio by one while using only a fixed small lower bound on the bucket count b . These two losses cannot both be worst-case, and

retaining their correlation improves both universal constants (Section 3). Second, it studies what SparseStack gives for Nyström approximation of a PSD matrix, where the natural quantity is not an embedding of a fixed subspace but the whole Gram error $Z = \Pi^\top \Pi - I_n$ (Sections 4 and 5). Third, it treats sketch-dependent targets: a frame selected after seeing the sketch by randomly pivoted Cholesky (RP), and RP run directly on the sketched Gram matrix (Sections 6 and 7).

1.1 What improves, and what is inherited

The central contribution is a sharper *sufficient* parameter prescription for the same real, fully independent sketch. Its operator estimates are inherited and stated explicitly in Section 2; the new proof starts at the scalar envelope. The separate Nyström, drift, and selection results describe what can be deduced from the sketch’s moment structure. They make no claim of a new sketching distribution or of an asymptotically stronger embedding theorem.

Theorem A is the constants result. Theorems B–D give its approximation and selection companions. Theorem E records an application proved in another paper in this repository, so that readers can see the precise boundary between this paper and that dependency.

Throughout, $d \geq 1$, $0 < \varepsilon \leq 1$, $0 < \delta \leq 1$,

$$q = \max\{1, \lceil \log_4(d/\delta) \rceil\}, \quad v = 2q + 1, \quad D = d + v.$$

Theorem A (sharper universal constants). Let (C, B) be one of $(69, 8)$, $(74, 15/2)$, $(79, 7)$, and choose

$$s = \left\lceil \frac{Bv}{\varepsilon} \right\rceil, \quad b = \left\lceil \frac{CD}{s\varepsilon^2} \right\rceil, \quad m = sb.$$

Then for every n and every $U \in \mathbb{R}^{n \times d}$ with $U^\top U = I_d$, $\mathbb{P}\{\|U^\top \Pi^\top \Pi U - I_d\| > \varepsilon\} \leq \delta$. Moreover

$$s < \frac{(3B+1)q}{\varepsilon}, \quad m < \frac{Cd + (2C+2B)q + C + B + 1}{\varepsilon^2}.$$

In particular, the simpler universal row bound is

$$m < \frac{(2C+2B)(d+q)}{\varepsilon^2},$$

so the constant pairs $(3B+1, 2C+2B)$ are $(25, 154)$, $(23.5, 163)$ and $(22, 172)$, against $(26.5, 179)$ in [9].

This is Theorem 3.1. It uses the imported operator estimates of [9] unchanged; the new ingredient is a family of 21 interval envelopes, certified in exact rational arithmetic (Table 1 and Appendix A).

Theorem B (whole-sketch Nyström bound). Let $A \in \mathbb{R}^{n \times n}$ be PSD, $Z = \Pi^\top \Pi - I_n$, and $R = A - A\Pi^\top(\Pi A \Pi^\top)^+ \Pi A$. For every matrix Π , $\|R\|_F^2 \leq \text{tr}(AZAZ)$. For SparseStack,

$$\mathbb{E} \text{tr}(AZAZ) = \frac{(\text{tr } A)^2 + \|A\|_F^2 - 2 \sum_i A_{ii}^2}{m} \leq \frac{2(\text{tr } A)^2}{m}.$$

This combines Theorems 4.1 and 4.2. The bound counts the actual rows m and is blind to rank: it is exactly the variance of approximate matrix multiplication for $A^{1/2}A^{1/2}$. Proposition 4.4 shows that no finite-support oblivious sketch with $m < n$ can offer a uniform bound of the form $\mathbb{E}\|R\| \leq K \cdot (\text{tail of } A)$.

Theorem C (block drift and diagonal training). Whole independent CountSketch blocks satisfy the size-biased drift inequality with constant $\kappa_b = 1 + 2/b$; for every convex potential f with $f(0) = 0$, one fresh block decreases $\mathbb{E} \operatorname{tr} f(R)$ by at least $\operatorname{tr}[Rf(R)]/(\kappa_b \operatorname{tr} R)$. For $b \geq 3$ the sharp uniform order-one constant of a single normalized bucket row is exactly b . SparseStack also gives an unbiased diagonal estimator with $\mathbb{E} \|h - \operatorname{diag} R\|_2^2 = (\|R\|_F^2 - \sum_i R_{ii}^2)/m$.

See Theorem 5.3, Proposition 5.6 and Theorem 5.7. The block bounds credit a b -column block like a single probe; we quantify the resulting loss, which reaches a factor of about b .

Theorem D (adaptive frame selection). Fix an isometry $V \in \mathbb{C}^{N \times k}$ and targets U_J for all k -subsets J . If J is chosen by k steps of RP on $VC(S)V^*$ for a sketch-dependent positive-definite $C(S)$, then the law of J is at most $\Lambda_{C(S)} |\det V_J|^2$, where $1 \leq \Lambda_C = k! \det C / \prod_j \tau_j(C) \leq k!$. If every fixed target fails with probability at most δ_0 and $\Lambda_{C(S)} \leq \bar{\Lambda}$ almost surely, the selected target fails with probability at most $\bar{\Lambda} \delta_0$.

This is Proposition 6.1. Table 2 gives the SparseStack parameters: at $d = 100$, $\varepsilon = 1/2$, $\delta = 0.01$, $k = 10$, the baseline certificate needs $q = 18$ and 44,659 rows for the factorial gate (the new (69, 8) prescription needs 37,888), whereas a union bound over $\binom{1000}{10}$ frames needs $q = 46$ and 63,240 rows and a Rényi-divergence price needs $q = 72$ and 81,345 rows.

Theorem E (sketched RPCholesky, imported). Let $A = F^\top F$, fix a target rank r , and put $\eta_A = \sum_{j>r} \lambda_j(A) / \operatorname{tr} A > 0$. Randomly pivoted Cholesky run for k steps on the sketched Gram $(\Pi F)^\top (\Pi F)$ needs only embeddings of the $(k+1)$ -dimensional frames met along its own pivot path. For a sketch accuracy $\eta \leq 1/4$ and a target accuracy $\varepsilon \leq 1$, a SparseStack with $m = O((k + \log(1/(\varepsilon \eta_A)))/\eta^2)$ rows, independent of n and rank F , and $k = O(r(1/\varepsilon + \log(1/\varepsilon) + \log(1/\eta_A)))$ steps give $\mathbb{E} \operatorname{tr} R_A(J_k) \leq (1 + 2\varepsilon) \sum_{j>r} \lambda_j(A)$.

This is the SparseStack trace-recurrence consequence in Imported Theorem 7.2. Its path comparison and hybrid argument come from *RPCholesky under approximate and sketched pivot probabilities* [8]; we prove the SparseStack moment input (Lemma 7.1) and explain the specialization below.

1.2 Roadmap

Figure 1 shows how the results depend on the imported theorems. Section 2 recalls the model and states the imported estimates. Sections 3–7 contain the proofs. Section 8 collects numerical diagnostics, and Section 9 lists open problems.

1.3 Relation to existing work and contribution

Embedding constants. CountSketch originates with Charikar, Chen and Farach-Colton [2]. The block construction used here is that of Kane and Nelson [16] and one of the OSNAP variants of Nelson and Nguyen [17]. The name SparseStack and the subspace-injection viewpoint follow Camaño, Epperly, Meyer and Tropp [1]. Optimal dimension with near-optimal sparsity was obtained by Chenakkod, Dereziński and Dong [12]. The target dependence on d, ε, δ has also been proved in the SparseStack paper [10] and by Mai and Rao [11]. This companion’s contribution is the explicit finite certificate for the three constant pairs in Theorem A, relative to the direct certificate of [9, Theorem 10.6]. It does not claim priority for the construction or for these asymptotic rates.

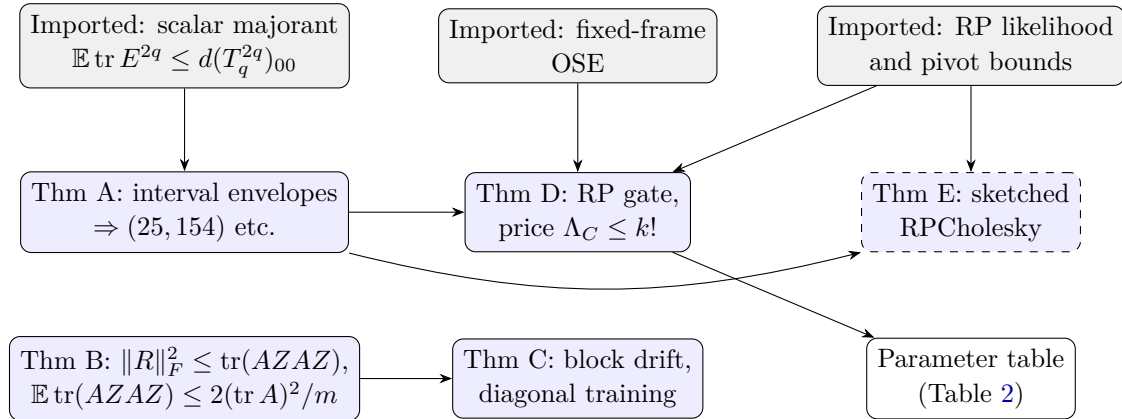


Figure 1: Dependencies. Grey boxes are imported black boxes; blue boxes are proved here. The dashed box (Theorem E) is proved in [8]; here we supply its SparseStack moment input from Theorem A. Theorem B and Theorem C use only second and fourth moments of the signed hash variables.

Approximation and estimation. The Nyström projector representation is standard, and the approximate-matrix-multiplication moment identity is part of sparse sketching [4, 1]. We give a short compression argument connecting the full Gram error to the squared Frobenius Nyström residual, together with the exact signed-hash variance and a finite-support obstruction. These are elementary refinements and applications of existing mechanisms, rather than a new tail-relative Nyström theory. The block ratio argument is inherited from the repository companion [7]; here its exact mixed denominator yields $\kappa_b = 1 + 2/b$. Stochastic trace and diagonal probing precede this work [13, 14], and variance-reduced trace and diagonal estimators include XTrace and XDiag [15]. Our estimation results identify the exact SparseStack variance and the conditionally unbiased correction from independently sampled diagonals; we do not claim the probing or control-variate ideas themselves.

Selection after observing the sketch. Randomly pivoted Cholesky is due to Chen, Epperly, Tropp and Webber [3]. The factorial adaptive-versus-volume comparison is Deshpande and Vempala’s Proposition 7 [5]; its proof already uses residual spectral-tail denominators. We retain those denominators as the explicit spectral factor Λ_C , and apply the classical comparison to transfer fixed-frame guarantees after a sketch-dependent selection. Thus the new application is the stated sketch-reuse guarantee and its parameter comparison, not a new factorial sampling theorem. The fractional likelihood bound is imported from [6], and the path-local sketched-RP theorem is imported from [8]. All repository dependencies are identified by their statements and linked in the bibliography. A literature check supports these bounded contribution claims; it is not an exhaustive claim of first occurrence.

2 The SparseStack model and the imported estimates

Definition 2.1 (SparseStack). Fix integers $s \geq 1$ and $b \geq 2$ and put $m = sb$. For $g \in [s]$ and $i \in [n]$ draw a bucket $h_g(i)$ uniformly from $[b]$ and a Rademacher sign σ_{gi} , all $2sn$ variables independent. The CountSketch block $H_g \in \mathbb{R}^{b \times n}$ has entries $(H_g)_{ai} = \sigma_{gi} \mathbf{1}\{h_g(i) = a\}$, and

$$\Pi = s^{-1/2} \begin{bmatrix} H_1^\top & \dots & H_s^\top \end{bmatrix}^\top \in \mathbb{R}^{m \times n}.$$

A direct computation gives the Gram matrix

$$(\Pi^\top \Pi)_{ij} = \frac{1}{s} \sum_{g=1}^s \sigma_{gi} \sigma_{gj} \mathbf{1}\{h_g(i) = h_g(j)\}, \quad (\Pi^\top \Pi)_{ii} = 1. \quad (1)$$

This representation drives Sections 4 and 5. For a fixed isometry $U \in \mathbb{R}^{n \times d}$ write $E = U^\top \Pi^\top \Pi U - I_d$. Put $h = b - 1$ and $r_b = b/(b - 1)$.

2.1 Imported theorems

We use two results of [9] as black boxes. Their exact source is Theorem 10.6 and Appendix B.4–B.5 of the repository framework, not the earlier coupled theorem in `SparseStack.pdf` [10]. That earlier paper has a different parameter prescription and a formal-verification statement for its own theorem. No formal verification is asserted for the sharper constants proved here.

Imported theorem 2.2 (Fixed-frame OSE [9, Theorem 10.6]). *Let q, v, D be as in Section 1.1, $s = \lceil (17/2)v/\varepsilon \rceil$, $b = \lceil 81D/(s\varepsilon^2) \rceil$ and $m = sb$. Then for every isometry $U \in \mathbb{R}^{n \times d}$, $\mathbb{P}\{\|E\| > \varepsilon\} \leq \delta$. Moreover $s < 26.5q/\varepsilon$ and $m < 179(d + q)/\varepsilon^2$.*

The second import is the operator part of the proof of Imported Theorem 2.2. For even grades $\nu \geq 2$ define

$$m\alpha_\nu = \sqrt{(m + \nu)(d + \nu + 1)} + \sqrt{h\nu(d + \nu - 1)/2} + \nu\sqrt{h} + \sqrt{bh\nu(\nu - 1)/2}, \quad (2)$$

$$m\chi_\nu = d + 2\nu - 1 + 2\sqrt{b(\nu - 1)(d + \nu - 1)} + (\nu + 1)\sqrt{b} + h\nu, \quad (3)$$

and $\alpha_0 = \sqrt{(d + 1)/m}$, $\chi_0 = 0$. Let T_q be the symmetric tridiagonal matrix indexed by $0, \dots, q$ with diagonal entries $(T_q)_{kk} = \chi_{2k}$ and off-diagonal entries $(T_q)_{k,k+1} = (T_q)_{k+1,k} = \alpha_{2k}$.

Imported theorem 2.3 (Scalar moment majorant [9, Appendix B.4, equations (224)–(226)]). *For all integers $n, d, q, s \geq 1$, $b \geq 2$ and every isometry $U \in \mathbb{R}^{n \times d}$,*

$$\mathbb{E} \operatorname{tr} E^{2q} \leq d (T_q^{2q})_{00}.$$

The sequence $\nu \mapsto \alpha_\nu$ is increasing on even $\nu \geq 0$.

In [9], the bound comes from an exact multiplication operator on the signed-bucket probability space, a vacuum moment identity, a shared-factor light-sector Gram estimate, and block Cauchy–Schwarz bounds on eight families of local legs; the numbers α_ν, χ_ν bound the grade-raising and grade-preserving parts of that operator. Nothing in the present paper reopens these estimates. The next two lemmas are the elementary scalar steps of [9]; we include their short proofs because Theorem A applies them to a new family of envelopes.

Lemma 2.4 (Triangular row-sum product). *Let T be a nonnegative symmetric tridiagonal matrix indexed by $0, \dots, q$ with $T_{00} = 0$ and $T_{01} \leq a$. Suppose its row sums are bounded by an increasing sequence $R_0 \leq R_1 \leq \dots \leq R_q$. For $q \geq 2$,*

$$(T^{2q})_{00} \leq a^2 R_1 R_q \prod_{k=2}^{q-1} R_k^2.$$

Proof. A nonzero returning walk of length $2q$ starts with the step $0 \rightarrow 1$ and ends with $1 \rightarrow 0$; these two steps have total weight at most a^2 . At time t such a walk sits at an index at most $\min(t, 2q - t)$. Restrict the intermediate states to this range; no contributing walk is lost. At departure times $t = 1, \dots, 2q - 2$ the largest permitted row sums are $R_1, R_2, \dots, R_q, R_{q-1}, \dots, R_2$. Applying the row-sum bounds successively to the restricted transition matrices, starting from the indicator of the final state, bounds the total weight by the product of these row sums. $\square$

Lemma 2.5 (Geometric-mean envelope). *Let T be as in Lemma 2.4, and let Ψ be positive, increasing and concave on $[0, 1]$. Suppose that, for $0 \leq k \leq q$, the k -th row sum of T/ε is at most $\Psi(u_k)$ with $u_k = (2k + 1)/(2q + 1)$, and $T_{01}/\varepsilon \leq a$. Put $J = \exp \int_0^1 \log \Psi(u) du$. If $a^2 J \leq \Psi(0)^3$ then, for $q \geq 2$,*

$$(T^{2q})_{00} \leq (\varepsilon J)^{2q}.$$

Moreover, for every $r \geq 1$,

$$J \leq \left[\prod_{j=0}^{r-1} \Psi\left(\frac{2j+1}{2r}\right) \right]^{1/r}. \quad (4)$$

Proof. If $T_{01} = 0$ the claim is trivial. Otherwise put $f = \log \Psi$, which is increasing and concave. Lemma 2.4 with $R_k = \varepsilon \Psi(u_k)$ gives

$$\frac{1}{2} \log \frac{(T^{2q})_{00}}{\varepsilon^{2q}} \leq \log a + \frac{1}{2} f(u_1) + \sum_{k=2}^{q-1} f(u_k) + \frac{1}{2} f(1).$$

The nodes $u_1 < \dots < u_q = 1$ have spacing $2/(2q + 1)$, so the trapezoidal sum on the right is at most $(q + \frac{1}{2}) \int_{u_1}^1 f$ by concavity. Since $(q + \frac{1}{2})u_1 = \frac{3}{2}$ and $f \geq f(0)$, this is at most $q \log J + \frac{1}{2} \log J - \frac{3}{2} \log \Psi(0)$. The boundary condition makes $\log a + \frac{1}{2} \log J - \frac{3}{2} \log \Psi(0) \leq 0$, and exponentiation proves the first claim. For (4), apply Jensen's inequality to the concave function f on each of the r intervals $[j/r, (j+1)/r]$. $\square$

For comparison, [9] applies Lemma 2.5 to one envelope with $(C, B) = (81, 17/2)$, a fixed lower bound $b \geq 10$, and the dimension ratio replaced by one; a four-node midpoint product then certifies $J < 1/2$. The next section refines exactly this last step.

3 Sharper universal constants

Theorem 3.1 (Refined universal constants). *Let $(C, B) \in \{(69, 8), (74, 15/2), (79, 7)\}$. With*

$$s = \left\lceil \frac{Bv}{\varepsilon} \right\rceil, \quad b = \left\lceil \frac{CD}{s\varepsilon^2} \right\rceil, \quad m = sb,$$

the *SparseStack* matrix satisfies, for every n and every isometry $U \in \mathbb{R}^{n \times d}$,

$$\mathbb{P}\{\|U^\top \Pi^\top \Pi U - I_d\| > \varepsilon\} \leq \delta.$$

The parameters obey

$$s < \frac{2Bq + B + 1}{\varepsilon} \leq \frac{(3B + 1)q}{\varepsilon}, \quad (5)$$

$$m < \frac{Cd + (2C + 2B)q + C + B + 1}{\varepsilon^2} < \frac{(2C + 2B)(d + q)}{\varepsilon^2}. \quad (6)$$

Concretely, the three-term row bound (6) reads

$$\varepsilon^2 m < 69d + 154q + 78, \quad 74d + 163q + 82.5, \quad 79d + 172q + 87,$$

against $81d + 179q + 90.5$ for the prescription of [9]. The pair $(69, 8)$ minimizes the row constant, $(79, 7)$ minimizes the sparsity constant, and all three improve both constants of $(26.5, 179)$. The coefficient of d in the three-term form is C , not $2C + 2B$; in the regime $d \gg q$ this is the relevant constant, and it drops from 81 to 69.

The proof occupies the rest of this section. It changes nothing in Imported Theorem 2.3; only the scalar bound on $(T_q^{2q})_{00}$ is new.

3.1 Retaining the dimension–bucket relation

Assume $q \geq 2$, so $v \geq 5$. Put

$$t = \frac{v}{D} \in (0, 1), \quad u = \frac{\nu + 1}{v}, \quad \theta_t(u) = 1 - t + tu = \frac{d + \nu + 1}{D}.$$

The prescription gives $m \geq CD/\varepsilon^2$ and $s \geq Bv/\varepsilon$.

Lemma 3.2 (Bucket lower bounds). *For $q \geq 2$,*

$$b > \frac{C}{B} \quad \text{and} \quad b > \frac{C}{t(B + 1/5)}. \quad (7)$$

Proof. From $s < (Bv + \varepsilon)/\varepsilon$,

$$b \geq \frac{CD}{s\varepsilon^2} > \frac{CD}{\varepsilon(Bv + \varepsilon)}.$$

Since $Bd > 1 \geq \varepsilon$, we have $BD > Bv + \varepsilon$, so the right side exceeds $C/(B\varepsilon) \geq C/B$. Also $D = v/t$, so the right side equals $C/(t\varepsilon(B + \varepsilon/v)) \geq C/(t(B + 1/5))$, because $\varepsilon \leq 1$ and $v \geq 5$. $\square$

Write $G_b = (2\sqrt{1 - 1/b} + 1)/\sqrt{b}$. With $y = b^{-1/2}$, $G_b = 2y\sqrt{1 - y^2} + y$, whose y -derivative $2(1 - 2y^2)/\sqrt{1 - y^2} + 1$ is positive on $[0, 1/\sqrt{2}]$. Hence G_b decreases in $b \geq 2$.

Lemma 3.3 (Parameter envelope). *For $q \geq 2$ and $1 \leq k \leq q$, the k -th row sum of T_q/ε is at most $\Phi_{t,b}(u_k)$, $u_k = (2k + 1)/v$, where*

$$\Phi_{t,b}(u) = \frac{2}{\sqrt{C}} \sqrt{\left(1 + \frac{tu}{C}\right)\theta_t(u)} + \frac{2 + \sqrt{2}}{\sqrt{CB}} \sqrt{u\theta_t(u)} + \frac{\theta_t(u)}{C} + \frac{1 + \sqrt{2} + G_b}{B} u. \quad (8)$$

The row $k = 0$ has sum α_0/ε , which is at most half the first term of $\Phi_{t,b}(u_0)$.

Proof. Fix $\nu = 2k \geq 2$ and $u = (\nu + 1)/\nu$. Since α increases, the k -th row sum of T_q is at most $2\alpha_\nu + \chi_\nu$. We use

$$\frac{d + \nu + 1}{m} \leq \frac{\varepsilon^2 \theta_t(u)}{C}, \quad \frac{\nu}{m} \leq \frac{\nu}{CD} \leq \frac{tu}{C}, \quad \frac{h}{m} \leq \frac{b}{m} = \frac{1}{s} \leq \frac{\varepsilon}{Bv},$$

together with $\varepsilon \leq 1$ to remove leftover factors $\sqrt{\varepsilon}$ and ε . Each group of terms in $(2\alpha_\nu + \chi_\nu)/\varepsilon$ is bounded as follows.

1. Light creation.

$$\frac{2\sqrt{(m + \nu)(d + \nu + 1)}}{m\varepsilon} = \frac{2}{\varepsilon} \sqrt{\left(1 + \frac{\nu}{m}\right) \frac{d + \nu + 1}{m}} \leq \frac{2}{\sqrt{C}} \sqrt{\left(1 + \frac{tu}{C}\right) \theta_t(u)}.$$

2. Mixed terms. From α_ν and from χ_ν , respectively,

$$\begin{aligned} \frac{2\sqrt{h\nu(d + \nu - 1)/2}}{m\varepsilon} &\leq \frac{2}{\varepsilon} \sqrt{\frac{\varepsilon\nu}{2Bv} \cdot \frac{\varepsilon^2 \theta_t}{C}} \leq \sqrt{2} \sqrt{\frac{u\theta_t}{CB}}, \\ \frac{2\sqrt{b(\nu - 1)(d + \nu - 1)}}{m\varepsilon} &\leq \frac{2}{\varepsilon} \sqrt{\frac{\varepsilon\nu}{Bv} \cdot \frac{\varepsilon^2 \theta_t}{C}} \leq 2\sqrt{\frac{u\theta_t}{CB}}. \end{aligned}$$

3. Heavy creation. $2\sqrt{bh\nu(\nu - 1)/2}/(m\varepsilon) \leq \sqrt{2}b\nu/(m\varepsilon) = \sqrt{2}\nu/(s\varepsilon) \leq \sqrt{2}u/B$.

4. Nonradical loops. $d + 2\nu - 1 + h\nu = (d + \nu + 1) + (b\nu - 2)$, so these terms contribute at most $\varepsilon\theta_t/C + \nu/(s\varepsilon) \leq \theta_t/C + u/B$.

5. Remaining terms. $2\nu\sqrt{h} + (\nu + 1)\sqrt{b} \leq (\nu + 1)bG_b$, so they contribute at most $(\nu + 1)G_b/(s\varepsilon) \leq G_b u/B$.

Summing gives (8). For $k = 0$ the row sum is $\alpha_0/\varepsilon = \varepsilon^{-1}\sqrt{(d + 1)/m} \leq \sqrt{\theta_t(1/v)/C}$. $\square$

The envelope of [9] is obtained by replacing θ_t by 1 and G_b by G_{10} . Those two losses cannot both be worst-case: when t is small, θ_t is close to one but Lemma 3.2 forces b to be large, and when t is large, θ_t is visibly smaller than one. Retaining this correlation is the entire source of the improvement.

3.2 A finite family of concave envelopes

Partition $(0, 1)$ by the 21 intervals

$$[\ell, r] = \left[\frac{j}{40}, \frac{j+1}{40}\right] \quad (0 \leq j \leq 19), \quad [\ell, r] = \left[\frac{1}{2}, 1\right].$$

For each interval put

$$n_{\ell,r} = \max\left\{\left\lfloor \frac{C}{B} \right\rfloor + 1, \left\lfloor \frac{C}{r(B + 1/5)} \right\rfloor + 1\right\}, \quad \vartheta_\ell(u) = 1 - \ell + \ell u, \quad (9)$$

and define

$$\Psi_{\ell,r}(u) = 2\sqrt{\frac{(1 + ru/C)\vartheta_\ell(u)}{C}} + (2 + \sqrt{2})\sqrt{\frac{u\vartheta_\ell(u)}{CB}} + \frac{\vartheta_\ell(u)}{C} + \frac{1 + \sqrt{2} + G_{n_{\ell,r}}}{B}u. \quad (10)$$

Lemma 3.4 (Interval envelopes). *Let $q \geq 2$ and $t = v/D \in [\ell, r]$. Then $b \geq n_{\ell,r}$ and $\Phi_{t,b} \leq \Psi_{\ell,r}$ on $[0, 1]$. Each $\Psi_{\ell,r}$ is positive, increasing and concave on $[0, 1]$, and the row sums of T_q/ε are at most $\Psi_{\ell,r}(u_k)$ for $0 \leq k \leq q$.*

Proof. Since b is an integer, Lemma 3.2 and $t \leq r$ give $b \geq \lfloor C/B \rfloor + 1$ and $b \geq \lfloor C/(r(B + 1/5)) \rfloor + 1$, so $b \geq n_{\ell,r}$ and $G_b \leq G_{n_{\ell,r}}$. For $u \in [0, 1]$, $\theta_t(u) = 1 - t(1 - u) \leq 1 - \ell(1 - u) = \vartheta_\ell(u)$ and $tu \leq ru$, which gives $\Phi_{t,b} \leq \Psi_{\ell,r}$ termwise. The two radical terms are geometric means of two nonnegative affine increasing functions, hence concave and increasing; the other terms are affine and increasing. The row-sum claim for $k \geq 1$ is Lemma 3.3; for $k = 0$ the row sum is at most half of the first term of $\Psi_{\ell,r}(u_0)$. $\square$

With the eight midpoint nodes $u_j = (2j + 1)/16$, the midpoint inequality (4) gives

$$J_{\ell,r} := \exp \int_0^1 \log \Psi_{\ell,r}(u) du \leq \left[\prod_{j=0}^7 \Psi_{\ell,r}(u_j) \right]^{1/8}. \quad (11)$$

Proposition 3.5 (Exact certificate). *For each of the three pairs (C, B) and each of the 21 intervals, $\prod_{j=0}^7 \Psi_{\ell,r}(u_j) < 2^{-8}$. Consequently $J_{\ell,r} < 1/2$. Table 1 lists $n_{\ell,r}$ and integers $T_{\ell,r} < 500000$ with $\prod_j \Psi_{\ell,r}(u_j) \leq (T_{\ell,r}/10^6)^8$.*

Proof. Every quantity in (10) at a rational node is rational except the square roots. First write $G_n = (2\sqrt{(n-1)/n} + 1)\sqrt{1/n}$, so no radical appears in a denominator. Replace each square root $\sqrt{x}$ by the rational $\lceil 10^{12}\sqrt{x} \rceil / 10^{12}$, computed exactly by an integer square root; this can only increase each factor, since all coefficients are nonnegative. The resulting rational upper bound $P_{\ell,r}$ of the eight-factor product is compared with 2^{-8} , and $T_{\ell,r}$ is the least integer with $T_{\ell,r}^8 \geq 10^{48}P_{\ell,r}$, all in exact integer arithmetic. The program in Appendix A performs these 63 checks; its output is Table 1. $\square$

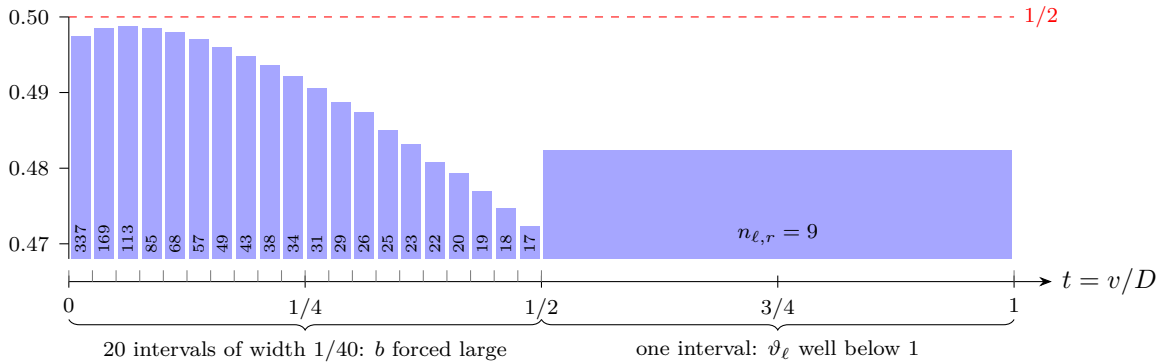


Figure 2: The interval family for $(C, B) = (69, 8)$. Bars show the certified geometric-mean bounds $T_{\ell,r}/10^6$ (axis from 0.468); numbers inside the bars are the forced bucket bounds $n_{\ell,r}$. Near $t = 0$ the dimension ratio ϑ_ℓ is close to one, but the buckets are numerous; near $t = 1$ buckets may be few, but ϑ_ℓ is small.

Interval $[\ell, r]$	(69, 8)		(74, 15/2)		(79, 7)	
	$n_{\ell,r}$	$T_{\ell,r}$	$n_{\ell,r}$	$T_{\ell,r}$	$n_{\ell,r}$	$T_{\ell,r}$
$[0, 1/40]$	337	497368	385	496334	439	497892
$[1/40, 1/20]$	169	498517	193	497515	220	499113
$[1/20, 3/40]$	113	498747	129	497786	147	499428
$[3/40, 1/10]$	85	498483	97	497567	110	499295
$[1/10, 1/8]$	68	497916	77	497122	88	498824
$[1/8, 3/20]$	57	497044	65	496231	74	498018
$[3/20, 7/40]$	49	496003	55	495421	63	497187
$[7/40, 1/5]$	43	494796	49	494096	55	496151
$[1/5, 9/40]$	38	493556	43	493000	49	494923
$[9/40, 1/4]$	34	492198	39	491521	44	493628
$[1/4, 11/40]$	31	490609	35	490208	40	492191
$[11/40, 3/10]$	29	488657	33	488206	37	490517
$[3/10, 13/40]$	26	487327	30	486727	34	488939
$[13/40, 7/20]$	25	484934	28	484852	32	486990
$[7/20, 3/8]$	23	483200	26	483043	30	485082
$[3/8, 2/5]$	22	480825	25	480650	28	483226
$[2/5, 17/40]$	20	479259	23	478951	26	481438
$[17/40, 9/20]$	19	476933	22	476572	25	479028
$[9/20, 19/40]$	18	474615	21	474184	24	476599
$[19/40, 1/2]$	17	472308	20	471789	22	474954
$[1/2, 1]$	9	482285	10	483066	12	484537

Table 1: Exact certificate for Theorem 3.1: bucket lower bounds $n_{\ell,r}$ and certified upper bounds $T_{\ell,r}/10^6$ for the eight-node geometric means $[\prod_j \Psi_{\ell,r}(u_j)]^{1/8}$. Every entry $T_{\ell,r}$ is below 500000; the largest is 499428, for (79, 7) on $[1/20, 3/40]$.

3.3 The vacuum boundary condition

Let $t \in [\ell, r]$ and let $a = \alpha_0/\varepsilon$ be the normalized vacuum edge. Then

$$a^2 \leq \frac{d+1}{CD} = \frac{1-t+t/v}{C} \leq \frac{1-4t/5}{C} \leq \frac{1-4\ell/5}{C},$$

using $v \geq 5$. Put $x = 1 - \ell \in [1/2, 1]$, so $1 - 4\ell/5 = (1 + 4x)/5$. Since $J_{\ell,r} < 1/2$,

$$a^2 J_{\ell,r} < \frac{1+4x}{10C} \leq \frac{3x}{5C} < \frac{8x^{3/2}}{C^{3/2}} \leq \Psi_{\ell,r}(0)^3.$$

The second inequality is $1 + 4x \leq 6x$, i.e. $x \geq 1/2$. The third is $3\sqrt{C} < 40\sqrt{x}$; squaring, it suffices that $9C < 800$, which holds for $C \leq 79$. The last uses $\Psi_{\ell,r}(0) \geq 2\sqrt{\vartheta_\ell(0)/C} = 2\sqrt{x/C}$. Thus the boundary condition of Lemma 2.5 holds for every interval and every $q \geq 2$.

3.4 Proof of Theorem 3.1

Case $q \geq 2$. Let $t = v/D$ and pick an interval $[\ell, r]$ containing t . By Lemma 3.4, Section 3.3 and Lemma 2.5 with $\Psi = \Psi_{\ell, r}$, together with Proposition 3.5 and Imported Theorem 2.3,

$$\mathbb{E} \operatorname{tr} E^{2q} \leq d (T_q^{2q})_{00} \leq d (\varepsilon J_{\ell, r})^{2q} < d (\varepsilon/2)^{2q}.$$

Since $\|E\|^{2q} \leq \operatorname{tr} E^{2q}$, Markov's inequality gives $\mathbb{P}\{\|E\| > \varepsilon\} < d4^{-q} \leq \delta$.

Case $q = 1$. Here $(T_1^2)_{00} = \alpha_0^2 = (d+1)/m \leq \varepsilon^2(d+1)/(CD) \leq \varepsilon^2/C$, so $\mathbb{P}\{\|E\| > \varepsilon\} \leq \mathbb{E} \operatorname{tr} E^2 / \varepsilon^2 \leq d/C$. As $q = 1$ forces $d/\delta \leq 4$, we get $d/C \leq 4\delta/C \leq \delta$.

Rounding. Since $s < Bv/\varepsilon + 1 \leq (Bv+1)/\varepsilon$, $s < (2Bq+B+1)/\varepsilon \leq (3B+1)q/\varepsilon$, using $q \geq 1$. Next, $m = sb < s(CD/(s\varepsilon^2) + 1) = CD/\varepsilon^2 + s$, and with $\varepsilon \leq 1$,

$$\varepsilon^2 m < C(d+2q+1) + 2Bq + B + 1 = Cd + (2C+2B)q + C + B + 1.$$

Finally $Cd + C + B + 1 < (2C+2B)d$ because $(C+2B)d \geq C+2B > C+B+1$ for $B > 1$. This proves (5) and (6). $\square$

Remark 3.6 (Where the remaining slack lies). The interval envelopes recover about 6% of the sparsity constant and 14% of the row constant. A larger reserve sits in the scalar step itself: Lemma 2.4 bounds all returning walks by a product of row sums. Iterating the tridiagonal majorant T_q directly, with the prescription of Theorem 3.1, gives $(T_q^{2q})_{00}^{1/(2q)}/\varepsilon \approx 0.35$ at $(C, B) = (69, 8)$ over a grid reaching $d = 10^9$ and $q = 3000$, far below the value $1/2$ that the proof needs; with $(C, B) = (35, 8)$ the same diagnostic stays below 0.48 (Section 8.1). This suggests the following.

Conjecture 3.7. *With $B = 8$ and $C = 35$, the prescription of Theorem 3.1 satisfies $(T_q^{2q})_{00} \leq (\varepsilon/2)^{2q}$ for all $d \geq 1$, $q \geq 2$ and $0 < \varepsilon \leq 1$. Consequently SparseStack would be an OSE with $m < (35d + 86q + 44)/\varepsilon^2$ rows.*

A proof would need a sharper bound on returning walks than Lemma 2.4, for example a diagonal similarity or a Perron analysis of the continuum limit of T_q , certified in exact arithmetic over a partition in t as above.

4 A whole-sketch Nyström Frobenius bound

For a PSD matrix $A \in \mathbb{R}^{n \times n}$ and any $\Pi \in \mathbb{R}^{m \times n}$, the Nyström approximation and its residual are

$$\mathcal{N}_A(\Pi) = A\Pi^\top(\Pi A\Pi^\top)^+\Pi A, \quad R = A - \mathcal{N}_A(\Pi).$$

Write $Z = \Pi^\top \Pi - I_n$ for the Gram error of the whole sketch. Unlike an OSE, which controls Z on one d -dimensional subspace, the bounds of this section see all of Z , weighted by A .

Theorem 4.1 (Deterministic bound). *For every PSD A and every Π ,*

$$0 \preceq R \preceq A \quad \text{and} \quad \|R\|_F^2 \leq \operatorname{tr}(AZAZ) = \|A^{1/2}ZA^{1/2}\|_F^2.$$

Proof. Represent the residual as a compression. Let $X = A^{1/2}$ and $K = X\Pi^\top \in \mathbb{R}^{n \times m}$. The matrix $P = K(K^\top K)^+ K^\top$ is the orthogonal projector onto range K , and $K^\top K = \Pi A \Pi^\top$, $XK = A\Pi^\top$. Hence $XPX = \mathcal{N}_A(\Pi)$ and

$$R = XQX, \quad Q = I - P.$$

This shows $0 \preceq R \preceq A$. Since $Q^2 = Q$,

$$\|R\|_F^2 = \text{tr}(XQX^2QX) = \text{tr}(QAAQ) = \text{tr}(QAAQ \cdot QAAQ) = \|QAAQ\|_F^2.$$

Express the missing part through the Gram error. Now $KK^\top = X\Pi^\top \Pi X = A + XZX$ has the same range as K , so $Q(A + XZX) = 0$, that is $QA = -QW$ with $W = XZX$ symmetric. Therefore $QAAQ = -QWQ$ and

$$\|R\|_F^2 = \|QWQ\|_F^2 \leq \|W\|_F^2 = \text{tr}(XZX XZX) = \text{tr}(AZAZ),$$

because an orthogonal projector is a contraction on both sides. $\square$

The identity $\|R\|_F = \|Q(XZX)Q\|_F$ inside the proof is exact; only the compression by Q is dropped. The right side $\|XZX\|_F^2 = \|X^\top \Pi^\top \Pi X - X^\top X\|_F^2$ is the squared error of sketched matrix multiplication for the product $X^\top X = A$.

4.1 The exact SparseStack computation

Theorem 4.2 (Exact second moment). *Let Π be SparseStack with $m = sb$ rows and let A be a fixed symmetric matrix. Then*

$$\mathbb{E} \text{tr}(AZAZ) = \frac{(\text{tr } A)^2 + \|A\|_F^2 - 2 \sum_i A_{ii}^2}{m}. \quad (12)$$

If $A \succeq 0$, then $\mathbb{E}\|R\|_F^2 \leq \mathbb{E} \text{tr}(AZAZ) \leq 2(\text{tr } A)^2/m$.

Proof. By (1), $Z_{ii} = 0$ and for $i \neq j$

$$Z_{ij} = \frac{1}{s} \sum_{g=1}^s \sigma_{gi} \sigma_{gj} \mathbf{1}\{h_g(i) = h_g(j)\}.$$

Compute the pair covariance. We claim that the off-diagonal entries have mean zero and variance $1/m$, and that entries indexed by distinct unordered pairs are uncorrelated. Indeed $\mathbb{E}Z_{ij} = 0$ because σ_{gi} is independent of all other variables in the g -th summand. For unordered pairs $\{i, j\} \neq \{k, l\}$ with $i \neq j, k \neq l$, consider a product $\sigma_{gi} \sigma_{gj} \sigma_{g'k} \sigma_{g'l}$. If $g \neq g'$ it has mean zero by independence of the blocks. If $g = g'$, some index appears exactly once among i, j, k, l , and the corresponding sign has mean zero and is independent of the rest, including the hashes. Hence $\mathbb{E}Z_{ij}Z_{kl} = 0$. Finally, cross terms between different blocks vanish, so

$$\mathbb{E}Z_{ij}^2 = \frac{1}{s^2} \sum_{g=1}^s \mathbb{P}\{h_g(i) = h_g(j)\} = \frac{1}{s^2} \cdot \frac{s}{b} = \frac{1}{m}.$$

Contract the covariance with the matrix. Expand the trace as

$$\text{tr}(AZAZ) = \sum_{i,j,k,l} A_{li} Z_{ij} A_{jk} Z_{kl}.$$

The expectation of $Z_{ij}Z_{kl}$ is nonzero only if $i \neq j$ and $(k, l) \in \{(i, j), (j, i)\}$. The choice $(k, l) = (j, i)$ contributes $A_{ii}A_{jj}/m$, and $(k, l) = (i, j)$ contributes A_{ij}^2/m . Summing over ordered pairs $i \neq j$,

$$\mathbb{E} \operatorname{tr}(AZAZ) = \frac{1}{m} \left[\sum_{i \neq j} A_{ii}A_{jj} + \sum_{i \neq j} A_{ij}^2 \right].$$

The first sum is $(\operatorname{tr} A)^2 - \sum_i A_{ii}^2$; the second is $\|A\|_F^2 - \sum_i A_{ii}^2$. Therefore

$$\mathbb{E} \operatorname{tr}(AZAZ) = \frac{(\operatorname{tr} A)^2 + \|A\|_F^2 - 2 \sum_i A_{ii}^2}{m}.$$

For $A \succeq 0$, $\|A\|_F^2 \leq (\operatorname{tr} A)^2$ gives the final bound, and Theorem 4.1 gives $\mathbb{E}\|R\|_F^2 \leq \mathbb{E} \operatorname{tr}(AZAZ)$. $\square$

Remark 4.3 (What the bound counts). The bound $2(\operatorname{tr} A)^2/m$ counts the actual rows $m = sb$ of the sketch, not the number of blocks, and it does not depend on s and b separately. It is blind to rank: it neither improves when A has low rank nor involves a tail $\sum_{j>r} \lambda_j(A)$. Only the second-order structure of Z enters the proof, so the same computation holds whenever the blocks are independent, the signs in each block are four-wise independent and independent of the hashes, and the hashes in each block are pairwise independent. The identity (12) is the classical sparse-sketch variance of approximate matrix multiplication for $A^{1/2}A^{1/2}$; a Gaussian sketch with m rows has the larger value $[(\operatorname{tr} A)^2 + \|A\|_F^2]/m$, since its diagonal Gram error does not vanish.

4.2 No uniform expected tail-relative guarantee

A natural wish is a bound $\mathbb{E}\|R\| \leq K\tau$, where τ is a tail of the spectrum of A . For a sketch with finitely many values, this is impossible whenever $m < n$.

Proposition 4.4 (Impossibility). *Let Π be a random $m \times n$ matrix with finite support and $m < n$, and let $1 \leq r < n$. For every $K < \infty$ there is a PSD A with*

$$\mathbb{E} \operatorname{tr} R > K \sum_{j>r} \lambda_j(A) \quad \text{and} \quad \mathbb{E}\|R\|_F^2 > K \left(\sum_{j>r} \lambda_j(A) \right)^2.$$

Proof. Pick Π_0 in the support, with $p_0 = \mathbb{P}\{\Pi = \Pi_0\} > 0$. Since $m < n$ there is a unit vector $u \in \ker \Pi_0$. For $H > 0$ put $A_H = H u u^\top + I_n$, whose eigenvalues are $H + 1, 1, \dots, 1$, so $\sum_{j>r} \lambda_j(A_H) = n - r$ does not depend on H . On the event $\Pi = \Pi_0$ we have $\Pi_0 A_H \Pi_0^\top = \Pi_0 \Pi_0^\top$ and $A_H \Pi_0^\top = \Pi_0^\top$, so $\mathcal{N}_{A_H}(\Pi_0) = P_0$, the orthogonal projector onto $\operatorname{range} \Pi_0^\top \perp u$. Hence $R = H u u^\top + (I - P_0) \succeq (H + 1) u u^\top$, and

$$\mathbb{E} \operatorname{tr} R \geq p_0(H + 1), \quad \mathbb{E}\|R\|_F^2 \geq p_0(H + 1)^2.$$

Letting $H \rightarrow \infty$ proves both claims. $\square$

SparseStack has finite support, and for the matrices A_H of the proof the damage is done on the event $\{\Pi u = 0\}$, whose probability is exponentially small in s for SparseStack (for $u = (e_1 - e_2)/\sqrt{2}$, say, it equals $(2b)^{-s}$). The proposition therefore rules out uniform *expected* tail-relative guarantees, but not high-probability guarantees: a failure budget δ exceeding the probability of the bad event absorbs it. For finite-support sketches the right targets are

high-probability statements, such as those obtained from an OSE for the leading eigenspace, or validated guarantees, in which the error is estimated a posteriori, for example by the independent diagonal and trace estimators of Section 5.4. Rank-blind expected bounds such as Theorem 4.2 remain possible because they do not compare with the tail.

5 Block drift, bucket rows, and diagonal training

This section treats sequential Nyström approximation with whole CountSketch blocks, using the size-biased ratio argument of *Product-Probe Drift, Kronecker Nyström Approximation, and Trace Estimation* [7]. Throughout, $H \in \mathbb{R}^{b \times n}$ is one CountSketch block with independent uniform buckets $h_i \in [b]$ and independent signs σ_i , and for a symmetric M we write $X_M = \text{tr}(HMH^\top)$.

5.1 Mixed moments and the block ratio

Lemma 5.1 (Mixed trace identity). *For symmetric $B, R \in \mathbb{R}^{n \times n}$,*

$$\mathbb{E}X_B = \text{tr } B, \quad \mathbb{E}X_B X_R = \text{tr } B \text{ tr } R + \frac{2}{b} \left(\text{tr}(BR) - \sum_i B_{ii} R_{ii} \right).$$

If $B, R \succeq 0$ then $\mathbb{E}X_B X_R \leq \kappa_b \text{tr } B \text{tr } R$ with $\kappa_b = 1 + 2/b$.

Proof. Expand $X_B = \text{tr } B + \sum_{i \neq j} B_{ij} \sigma_i \sigma_j \mathbf{1}\{h_i = h_j\}$. As in the proof of Theorem 4.2, distinct unordered pairs are uncorrelated and $\mathbb{E}(\sigma_i \sigma_j \mathbf{1}\{h_i = h_j\})^2 = 1/b$; the two orderings of a pair give the factor 2. For PSD matrices $B_{ii} R_{ii} \geq 0$ and $\text{tr}(BR) \leq \text{tr } B \text{tr } R$. $\square$

Lemma 5.2 (Block ratio). *Let $B, R \succeq 0$ with $\text{tr } B > 0$ and $\ker R \subseteq \ker B$, and set $X_B/X_R = 0$ when $X_R = 0$. Then*

$$\mathbb{E} \frac{X_B}{X_R} \geq \frac{(\text{tr } B)^2}{\mathbb{E}X_B X_R} \geq \frac{\text{tr } B}{\kappa_b \text{tr } R}.$$

Proof. If $X_R = 0$ then $R^{1/2}H^\top = 0$, so $\text{range } H^\top \subseteq \ker R \subseteq \ker B$ and $X_B = 0$. The useful change of measure gives more weight to sketches carrying more of B . Define $d\mathbb{Q} = X_B d\mathbb{P} / \text{tr } B$. Under $\mathbb{Q}$, $X_R > 0$ almost surely. Jensen's inequality for $x \mapsto 1/x$ now gives

$$\begin{aligned} \mathbb{E} \frac{X_B}{X_R} &= \text{tr } B \mathbb{E}_{\mathbb{Q}} \frac{1}{X_R} \\ &\geq \frac{\text{tr } B}{\mathbb{E}_{\mathbb{Q}} X_R} = \frac{(\text{tr } B)^2}{\mathbb{E}X_B X_R}. \end{aligned}$$

Lemma 5.1 bounds the denominator and finishes the proof. $\square$

5.2 Block drift and convex potentials

For $R \succeq 0$ the Nyström update by a fresh block H is $R' = R - RH^\top(HRH^\top)^+HR$. Updating successively by independent blocks $H_1, \dots, H_s$ from $R_0 = A$ yields $R_s = A - \mathcal{N}_A(Z_s)$ with $Z_s = [H_1^\top \cdots H_s^\top]^\top$, because the projector onto $\text{range}(A^{1/2}Z_s^\top)$ is the sum of the projector onto the previous range and the projector onto the new part of the range. The normalization $s^{-1/2}$ of SparseStack does not change the Nyström approximation.

Theorem 5.3 (Block drift). *For every nonzero $R \succeq 0$, $0 \preceq R' \preceq R$ and*

$$\mathbb{E}(\text{tr } R - \text{tr } R') \geq \frac{\text{tr}(R^2)}{\kappa_b \text{tr } R}.$$

More generally, for every convex $f : [0, \infty) \rightarrow [0, \infty)$ with $f(0) = 0$,

$$\mathbb{E} \text{tr } f(R') \leq \text{tr } f(R) - \frac{\text{tr}[Rf(R)]}{\kappa_b \text{tr } R}. \quad (13)$$

Proof. Lower-bound what the sampled range captures. Let $F = R^{1/2}H^\top H R^{1/2}$, and let P be the orthogonal projector onto $\text{range } F$, which equals $\text{range}(R^{1/2}H^\top)$. Then $R' = R^{1/2}(I - P)R^{1/2}$, so $0 \preceq R' \preceq R$ and $\text{tr } R - \text{tr } R' = \text{tr}(PR)$. Suppose $X_R = \text{tr } F > 0$. Every eigenvalue of F is at most $\text{tr } F$, so $F/X_R \preceq P$, and for every $M \succeq 0$ commuting with R ,

$$\text{tr}(PM) \geq \frac{\text{tr}(FM)}{X_R} = \frac{X_{R^{1/2}MR^{1/2}}}{X_R}.$$

With $M = R$ this is $\text{tr}(PR) \geq X_{R^2}/X_R$; with $M = f(R) \succeq 0$ it is $\text{tr}(Pf(R)) \geq X_{Rf(R)}/X_R$. Both inequalities also hold when $X_R = 0$.

Pass from captured mass to a convex potential. Next, R' has the same nonzero eigenvalues as the compression $(I - P)R(I - P)$. Let $w_1, w_2, \dots$ be orthonormal eigenvectors of this compression spanning $\text{range}(I - P)$, with eigenvalues $\mu_k = w_k^\top R w_k$. Jensen's inequality for the spectral measure of R in the state w_k gives $f(\mu_k) \leq w_k^\top f(R) w_k$, and $f(0) = 0$, so

$$\text{tr } f(R') \leq \sum_k w_k^\top f(R) w_k = \text{tr } f(R) - \text{tr}(Pf(R)).$$

Average the size-biased ratio. Apply Lemma 5.2 with $B = R^2$ or $B = Rf(R)$; both satisfy $\ker R \subseteq \ker B$. If $Rf(R) = 0$ the claim (13) is immediate. $\square$

Keeping the exact denominator in Lemma 5.2 gives the residual-dependent refinement

$$\mathbb{E}(\text{tr } R - \text{tr } R') \geq \frac{[\text{tr}(R^2)]^2}{\text{tr}(R^2) \text{tr } R + \frac{2}{b}(\text{tr}(R^3) - \sum_i (R^2)_{ii} R_{ii})}. \quad (14)$$

Corollary 5.4 (Independent-block Frobenius bound). *With $T = \text{tr } A > 0$ and s independent blocks, $\mathbb{E}\|R_s\|_F^2 \leq \kappa_b T^2/s$.*

Proof. Conditionally on R_t , Theorem 5.3 and $\text{tr } R_t \leq T$ give $\mathbb{E}[\text{tr } R_t - \text{tr } R_{t+1} \mid R_t] \geq \text{tr}(R_t^2)/(\kappa_b T)$. Summing, $\sum_{t \leq s} \mathbb{E} \text{tr}(R_t^2) \leq \kappa_b T \mathbb{E}(\text{tr } R_0 - \text{tr } R_s) \leq \kappa_b T^2$. Since $R_{t+1} \preceq R_t$, the sequence $\text{tr}(R_t^2)$ decreases, so $s \mathbb{E} \text{tr}(R_s^2) \leq \kappa_b T^2$. $\square$

Remark 5.5 (The block bounds lose up to a factor b). The proof of Theorem 5.3 uses $F \preceq (\text{tr } F)P$, which credits a b -column block with the progress of a single normalized probe. The bound is tight up to κ_b when the residual has rank one, but for residuals of rank at least b it underestimates the true progress by a factor of up to about b . Monte Carlo ratios of the observed mean trace drop to the bound $\text{tr}(R^2)/(\kappa_b \text{tr } R)$ are 4.96 for $R = I_{12}$, $b = 3$; 4.3 for a 12×12 Wishart matrix, $b = 3$; 9.95 for $R = I_{40}$, $b = 8$; and $1.67 = \kappa_3$ for a rank-one R (Table 3). For $A = I_n$, Corollary 5.4 is nontrivial only for $s > \kappa_b n$ blocks. In contrast, a single block captures exactly its number of occupied buckets, whose expectation is $b[1 - (1 - 1/b)^n]$.

This calculation exposes the progress lost when an entire block is credited as one probe; it does not assert guaranteed full rank at $s = \lceil n/b \rceil$. In the count of sketch columns, the block guarantees are therefore weaker than whole-sketch or per-column bounds: Theorem 4.2 gives $2T^2/m = 2T^2/(sb)$ where Corollary 5.4 gives $(b+2)T^2/(sb)$. What the block argument offers is the explicit constant κ_b , arbitrary convex potentials, and the use of independence only across blocks, never within a block.

5.3 The sharp order-one constant of a bucket row

Spherical probes enjoy entropy-type constants: for a uniformly random unit vector, $\mathbb{E}X^{1+\beta}$ grows slowly in β . A single row of a CountSketch block behaves very differently. Fix a bucket a and normalize the corresponding row to the isotropic vector $z_i = \sqrt{b}\sigma_i\mathbf{1}\{h_i = a\}$. For a unit vector u put $X_u = |u^\top z|^2$.

Proposition 5.6 (Bucket rows). *For every unit u , $\mathbb{E}X_u = 1$ and $\mathbb{E}X_u^2 = 3 + (b-3)\sum_i u_i^4$. If $b \geq 3$ then, for $1 \leq p \leq 2$,*

$$H(p) := \sup_{\|u\|=1} \mathbb{E}X_u^p = b^{p-1},$$

with equality at every coordinate vector $u = e_i$. Consequently the sharp uniform order-one constant is $\lim_{\beta \downarrow 0} H(1+\beta)^{1/\beta} = b = H(2)$; equivalently $\sup_{\|u\|=1} \mathbb{E}[X_u \log X_u] = \log b$.

Proof. The variables $\sigma_i\mathbf{1}\{h_i = a\}$ are independent, symmetric, with second moment $1/b$ and fourth moment $1/b$. Hence $\mathbb{E}X_u = \sum_i u_i^2 = 1$ and

$$\mathbb{E}X_u^2 = b^2 \left[\frac{1}{b} \sum_i u_i^4 + \frac{3}{b^2} \sum_{i \neq j} u_i^2 u_j^2 \right] = 3 + (b-3) \sum_i u_i^4 \leq b,$$

using $\sum_i u_i^4 \leq 1$ and $b \geq 3$. For $1 \leq p \leq 2$, Hölder's inequality gives $\mathbb{E}X^p = \mathbb{E}X^{2-p}X^{2(p-1)} \leq (\mathbb{E}X)^{2-p}(\mathbb{E}X^2)^{p-1} \leq b^{p-1}$. For $u = e_i$, $X = b$ with probability $1/b$ and $X = 0$ otherwise, so $\mathbb{E}X^p = b^{p-1}$. The limit follows. Finally $\mathbb{E}[X \log X] = \frac{d}{dp} \mathbb{E}X^p|_{p=1}$, and since $\mathbb{E}X_u^p \leq b^{p-1}$ with equality at $p = 1$, this derivative is at most $\log b$, with equality at $u = e_i$. $\square$

Thus the order-one constant of a bucket row equals its worst-case fourth moment, and spherical entropy constants cannot replace it row by row. This obstructs one particular transfer; it is not a lower bound on achievable embedding constants, nor on whole-block progress.

5.4 Diagonal training and trace estimation

Let $R \in \mathbb{R}^{n \times n}$ be symmetric and fixed, and let Π be an independent SparseStack with $m_0 = s_0 b$ rows. From the product $Y = R\Pi^\top$ form

$$h_i = \sum_{a=1}^{m_0} Y_{ia} \Pi_{ai} = (R\Pi^\top \Pi)_{ii}.$$

Theorem 5.7 (Diagonal and trace estimation). *The vector h is unbiased for $\text{diag } R$, and*

$$\mathbb{E}\|h - \text{diag } R\|_2^2 = \frac{\|R\|_F^2 - \sum_i R_{ii}^2}{m_0}.$$

The sketched trace $\sum_i h_i = \text{tr}(\Pi R \Pi^\top)$ is unbiased with $\text{Var}(\text{tr} \Pi R \Pi^\top) = 2(\|R\|_F^2 - \sum_i R_{ii}^2)/m_0$. For t independent uniform indices $I_1, \dots, I_t \in [n]$, also independent of Π , the corrected estimator

$$\hat{T}_R = \sum_i h_i + \frac{n}{t} \sum_{\ell=1}^t (R_{I_\ell I_\ell} - h_{I_\ell})$$

is unbiased for $\text{tr} R$ and $\text{Var} \hat{T}_R \leq \frac{n}{m_0 t} (\|R\|_F^2 - \sum_i R_{ii}^2)$.

Proof. With $Z = \Pi^\top \Pi - I$, $h_i - R_{ii} = (RZ)_{ii} = \sum_{j \neq i} R_{ij} Z_{ji}$. By the second-moment structure established in the proof of Theorem 4.2, the terms are uncorrelated with variances R_{ij}^2/m_0 ; summing over i gives the first formula. Next, $\text{tr}(RZ) = \sum_{i \neq j} R_{ij} Z_{ij} = 2 \sum_{i < j} R_{ij} Z_{ij}$, whose variance is $4 \sum_{i < j} R_{ij}^2/m_0 = 2(\|R\|_F^2 - \sum_i R_{ii}^2)/m_0$. For $\hat{T}_R$, condition on Π and write $e_i = R_{ii} - h_i$. The correction has conditional mean $\sum_i e_i$, so $\mathbb{E}[\hat{T}_R | \Pi] = \text{tr} R$ exactly, and

$$\text{Var} \hat{T}_R = \mathbb{E} \text{Var}(\hat{T}_R | \Pi) \leq \mathbb{E} \frac{n^2}{t} \cdot \frac{1}{n} \sum_i e_i^2 = \frac{n}{t} \mathbb{E} \|h - \text{diag} R\|_2^2. \quad \square$$

The sketched trace has exactly the variance of Hutchinson's estimator [13] with m_0 Rademacher probes, so at second order the sparse SparseStack probes lose nothing to dense ones. The corrected estimator serves a different purpose. Given the sketch, $\sum_i h_i$ is a fixed number that is off by $\text{tr}(RZ)$, whereas $\hat{T}_R$ is *conditionally* unbiased given Π , and its conditional variance can itself be estimated from the t sampled indices. This supplies a conditionally unbiased a-posteriori estimator for a residual such as $R = A - \mathcal{N}_A(\Pi_1)$, whose diagonal entries are cheap when $\text{diag} A$ is available. A certified confidence interval requires a separate concentration or sampling-error bound; conditional unbiasedness alone does not provide one. The training sketch must be independent of R : when R is a Nyström residual from an earlier sketch Π_1 , a fresh SparseStack is required. A known low-rank part of R can be added back exactly.

6 Adaptive frame selection by an RP gate

An OSE guarantee concerns a subspace fixed before the sketch is drawn. In applications the subspace is often chosen *after* looking at the sketch, and independence is lost. A union bound over a dictionary of N_{dict} candidate frames restores validity at the price of replacing δ by δ/N_{dict} . This section shows that when the choice is made by randomly pivoted Cholesky on a fixed frame, the price is a likelihood factor that never exceeds $k!$, however large the dictionary.

6.1 The gate

Randomly pivoted Cholesky (RP) on a PSD matrix M picks, at residual R , an index i with probability $R_{ii}/\text{tr} R$ and updates $R \leftarrow R - R_{:i} R_{i:}/R_{ii}$ [3].

Fix, before the sketch is drawn:

- an isometry $V \in \mathbb{C}^{N \times k}$;
- for every k -subset $J \subseteq [N]$ with $\det V_J \neq 0$, a target isometry U_J (for SparseStack, a real $n \times d$ isometry).

Here V_J is the $k \times k$ submatrix of rows indexed by J . Draw the sketch S and form a measurable positive-definite $k \times k$ matrix $C(S)$. Run k fresh RP steps on $VC(S)V^*$, holding $C(S)$ fixed, and release only the unordered set J of pivots. Finally apply the sketch S to the selected frame U_J . The reference law

$$\nu_V(J) = |\det V_J|^2$$

is a probability distribution by the Cauchy–Binet formula $\sum_J |\det V_J|^2 = \det(V^*V) = 1$, and it does not depend on S .

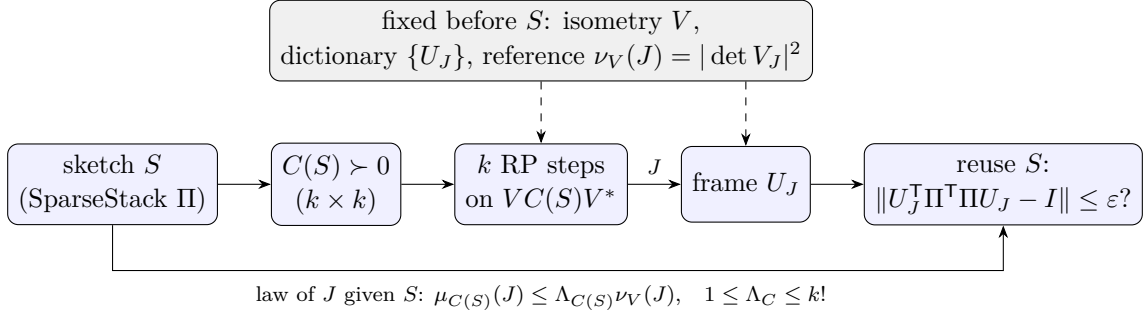


Figure 3: The gate pipeline. Only the grey data are fixed in advance; the frame is selected after the sketch is seen, and the same sketch is reused on the selected frame. The price of the dependence is a bound $\bar{\Lambda}$ on the likelihood ratio $\mu_{C(S)}/\nu_V$.

Proposition 6.1 (Factorial and spectral prices). *For the eigenvalues $\lambda_1 \geq \dots \geq \lambda_k > 0$ of C put*

$$\tau_j(C) = \sum_{i=j+1}^k \lambda_i \quad (0 \leq j < k), \quad \Lambda_C = \frac{k! \det C}{\prod_{j=0}^{k-1} \tau_j(C)}.$$

Conditionally on S , the released set has a law $\mu_{C(S)}$ with

$$\mu_{C(S)}(J) \leq \Lambda_{C(S)} \nu_V(J) \quad \text{for all } J, \quad 1 \leq \Lambda_C \leq k!.$$

Suppose that every fixed target U_J fails, under the law of S , with probability at most δ_0 , and that $\Lambda_{C(S)} \leq \bar{\Lambda}$ almost surely for a deterministic $\bar{\Lambda}$. Then the selected target U_J fails with probability at most $\bar{\Lambda}\delta_0$. In particular, $\delta_0 = \delta/k!$ always suffices.

Proof. Use the classical determinant comparison. Write $M = VCV^*$, of rank k . For an ordered pivot sequence $(i_1, \dots, i_k)$ with set J , the product of the pivot diagonals is the product of the Cholesky pivots of M_{JJ} , namely

$$\det M_{JJ} = \det(V_J C V_J^*) = \det C |\det V_J|^2.$$

After j steps, the removed matrix $M - R_j$ is PSD, of rank at most j , and satisfies $M - R_j \preceq M$. By Ky Fan's maximum principle its trace is at most $\lambda_1 + \dots + \lambda_j$, so $\text{tr } R_j \geq \tau_j(C)$. The probability of the ordered sequence is therefore at most $\det C \nu_V(J) / \prod_{j < k} \tau_j(C)$. Summing over the at most $k!$ orderings of J gives $\mu_C(J) \leq \Lambda_C \nu_V(J)$. Since $\lambda_{j+1} \leq \tau_j(C) \leq (k-j)\lambda_{j+1}$, we get $\det C \leq \prod_j \tau_j(C) \leq k! \det C$, i.e. $1 \leq \Lambda_C \leq k!$.

Transfer fixed-target failure probabilities. Let $\mathcal{B}_J$ be the failure event of the fixed target U_J , an event of S alone. The RP randomness is fresh given S , so

$$\mathbb{P}\{\mathcal{B}_J \text{ for the selected } J\} = \mathbb{E}_S \sum_J \mu_{C(S)}(J) \mathbf{1}_{\mathcal{B}_J}(S) \leq \bar{\Lambda} \sum_J \nu_V(J) \mathbb{P}_S(\mathcal{B}_J) \leq \bar{\Lambda} \delta_0.$$

The last step uses only fixed-target guarantees: the weights ν_V and the targets were fixed before S . $\square$

Lemma 6.2 (Spectral certificates). *If C is a multiple of I_k , then $\Lambda_C = 1$. If $C = I + \eta ww^*$ with $\|w\| = 1$ and $\eta \geq 0$, then $\Lambda_C = k(1+\eta)/(k+\eta)$. If the spectrum of C splits into contiguous bands of sizes n_ℓ whose largest-to-smallest ratios are at most ρ_ℓ , then*

$$\Lambda_C \leq \min\left\{k!, \frac{k!}{\prod_\ell n_\ell!} \prod_\ell \rho_\ell^{n_\ell}\right\};$$

in particular $\Lambda_C \leq k!/\prod_\ell n_\ell!$ when C has exact eigenvalue levels of multiplicities n_ℓ .

Proof. For $C = cI$, $\tau_j = (k-j)c$ and $\prod_j \tau_j = k! c^k = k! \det C$. For the spike, $\det C = 1 + \eta$, $\tau_0 = k + \eta$ and $\tau_j = k - j$ for $j \geq 1$, so $\Lambda_C = k!(1 + \eta)/((k + \eta)(k - 1)!)$. For bands, bound each τ_j from below by the eigenvalues remaining in its own band: if the j -th index is the i -th of band ℓ , then $\tau_j \geq (n_\ell - i + 1) \min_\ell$. Multiplying, $\prod_j \tau_j \geq \prod_\ell n_\ell! (\min_\ell)^{n_\ell}$, while $\det C \leq \prod_\ell (\max_\ell)^{n_\ell}$. $\square$

The gate needs a bound $\bar{\Lambda}$ that holds uniformly over the permitted values of $C(S)$. Inserting the observed value $\Lambda_{C(S)}$ into a failure budget chosen after seeing that same sketch is not justified by Proposition 6.1. The argument also does not cover a moving frame $V(S)$, a sketch-dependent dictionary, partial-rank stopping, or release of the pivot order.

6.2 SparseStack parameters after selection

Proposition 6.1 is distribution-free in S : it changes only the failure allowance of a fixed-frame certificate.

Corollary 6.3 (Selected-frame OSE). *In the setting of Proposition 6.1, let $S = \Pi$ be SparseStack and let the targets U_J be real $n \times d$ isometries. Replace q by*

$$q_{\bar{\Lambda}} = \max\{1, \lceil \log_4(d\bar{\Lambda}/\delta) \rceil\}$$

in either the prescription of Imported Theorem 2.2 or that of Theorem 3.1. Then $\mathbb{P}\{\|U_J^\top \Pi^\top \Pi U_J - I_d\| > \varepsilon\} \leq \delta$ for the selected J .

Proof. With $q_{\bar{\Lambda}}$, each fixed target fails with probability at most $d4^{-q_{\bar{\Lambda}}} \leq \delta/\bar{\Lambda}$ (for $q_{\bar{\Lambda}} = 1$, at most $d/C \leq \delta/\bar{\Lambda}$); apply Proposition 6.1. $\square$

For the price $k!$, the extra cost is $\log_4 k!$ in q , independent of the dictionary size. Table 2 gives exact integer parameters at $d = 100$, $\varepsilon = 1/2$, $\delta = 0.01$, $k = 10$ and $N = 1000$, for both prescriptions.

These are sufficient counts, not measured requirements. Spectral structure improves the price, but the elementary factorial price alone already beats the union bound by a wide margin.

	$(C, B) = (81, 17/2)$ [9]				$(C, B) = (69, 8)$, Thm. 3.1		
Selection price	q	s	b	m	s	b	m
Independent frame, $\overline{\Lambda} = 1$	7	255	147	37 485	240	133	31 920
Two exact levels $5 + 5$, $\overline{\Lambda} = 252$	11	391	102	39 882	368	93	34 224
Arbitrary spectrum, $\overline{\Lambda} = 10!$	18	629	71	44 659	592	64	37 888
Union bound, $\binom{1000}{10}$ frames	46	1581	40	63 240	1488	36	53 568
Rényi order $3/2$, $\alpha = \beta/(1 + \beta b_k)$	72	2465	33	81 345	2320	30	69 600
Rényi order $3/2$, optimal α	69	2363	33	77 979	2224	30	66 720
Rényi, optimized order (numerical)	32	1105	49	54 145	1040	44	45 760

Table 2: SparseStack parameters for a frame selected after the sketch, at $d = 100$, $\varepsilon = 1/2$, $\delta = 0.01$, $k = 10$, $N = 1000$. The factorial gate saves 29.4% of the rows and 60.2% of the nonzeros per column against the union bound; the two-level spectral gate saves more. The Rényi rows use Imported Theorem 6.4; see Section 6.3.

6.3 Why the factorial price beats the Rényi price

A different route compares μ_C with the volume law ν_V through a Rényi divergence. At full rank $k = \text{rank}(VCV^*)$, the volume-sampling law of VCV^* is exactly ν_V , because

$$\frac{\det(VCV^*)_{JJ}}{e_k(VCV^*)} = \frac{\det C |\det V_J|^2}{\det C}.$$

The following bound is proved in *Iterated-logarithmic pivot bounds for randomly pivoted Cholesky* [6]. For $k \geq 2$ and $0 < \beta < 1$ put $b_k = \log(1 + 2 \log(k!)/k)$, $a_k = k^k/k!$, and

$$B_{k,\beta}(\alpha) = kb_k\alpha - (k-1)\log \alpha, \quad 0 < \alpha \leq \beta.$$

The argument of [6] is valid for every $0 < \alpha \leq \beta$; the minimizing choice is $\alpha_{k,\beta} = \min\{\beta, (k-1)/(kb_k)\}$, and we write $B_{k,\beta} = B_{k,\beta}(\alpha_{k,\beta})$.

Imported theorem 6.4 (Fractional likelihood moment [6]). *Let $L = \mu/\nu$ be the likelihood ratio of the k -step RP set law to the volume law of a PSD matrix of rank at least k . For $p = 1/\beta > 1$,*

$$\mathbb{E}_\nu L^p \leq e^{pB_{k,\beta}} 2^k a_k^p.$$

The same bound holds with $B_{k,\beta}(\alpha)$ in place of $B_{k,\beta}$ for every $0 < \alpha \leq \beta$.

Hölder's inequality gives, for every event $\mathcal{E}$, $\mu(\mathcal{E}) \leq (\mathbb{E}_\nu L^p)^{1/p} \nu(\mathcal{E})^{1-1/p}$. Apply it conditionally on S to $\mathcal{E}_S = \{J : \text{the fixed target } U_J \text{ fails on } S\}$, and then average over S using the concavity of $x \mapsto x^{1-1/p}$ and $\mathbb{E}_S \nu_V(\mathcal{E}_S) \leq \delta_0$: the selected target fails with probability at most $(\sup_C \mathbb{E}_\nu L_C^p)^{1/p} \delta_0^{1-1/p}$. At $p = 3/2$ this is at most δ once

$$\log \frac{1}{\delta_0} = 3 \log \frac{1}{\delta} + \mathfrak{d}_k, \quad \mathfrak{d}_k = 3B_{k,2/3} + 2k \log 2 + 3 \log a_k.$$

The factorial gate instead needs $\log(1/\delta_0) = \log(1/\delta) + \log k!$. Both prices grow faster than linearly in k , but differently:

$$\log k! = k \log k - k + O(\log k), \quad \mathfrak{d}_k = k(6 + 2 \log 2 + 3 \log b_k) + O(\log k),$$

since $B_{k,\beta} = k(1 + \log b_k) + O(\log k)$ at the optimal α and $\log a_k = k - O(\log k)$. The Rényi price is linear in k up to the triply-logarithmic factor $\log b_k$, but with a large constant, and it also triples $\log(1/\delta)$. The factorial price wins as long as $\log k - 1 < 6 + 2 \log 2 + 3 \log b_k$, that is, while $\log k$ is below about 12. Exact evaluation at $\delta = 0.01$ places the crossover at $k \approx 1.33 \times 10^5$ with the optimal α and at $k \approx 1.66 \times 10^5$ with the simpler admissible choice $\alpha = \beta/(1 + \beta b_k)$. At $k = 10$ the Rényi price is $\log(1/\delta_0) \approx 90$ nats against 19.7 for the factorial price, which explains the last rows of Table 2.

Remark 6.5 (Optimizing the Rényi order). Hölder’s inequality works for every $p > 1$, with $\log(1/\delta_0) = \frac{p}{p-1} \log \frac{1}{\delta} + \frac{1}{p-1} \log \mathbb{E}_\nu L^p$. Numerically optimizing over p in Imported Theorem 6.4, the best order is near $p \approx 6.6$ at $k = 10$ and gives $q = 32$, still far worse than the factorial gate. With the order optimized separately for each k , the Rényi price drops below the factorial price from about $k \approx 130$ on (at $\delta = 0.01$). For selector ranks in the tens the elementary factorial gate is the right tool; for large k a likelihood-moment bound is preferable.

7 Sketched RPCholesky with SparseStack

Let $F \in \mathbb{R}^{p \times n}$ be a data matrix with Gram matrix $A = F^\top F$, fix a target rank $r \geq 1$, and put

$$\tau = \tau_r(A) = \sum_{j>r} \lambda_j(A) > 0, \quad \eta_A = \frac{\tau}{\text{tr } A}.$$

Sketched RP draws a SparseStack Π , runs k steps of RP on the sketched Gram $\hat{A} = (\Pi F)^\top (\Pi F)$, and returns the pivot set J_k together with the regression coefficients $\hat{X} = (\Pi F_J)^\dagger \Pi F$. The point is that $\hat{A}$ is formed from the $m \times n$ matrix ΠF , never from F itself. For a pivot set J write $R_A(J)$ and $R_{\hat{A}}(J)$ for the Schur-complement residuals of A and $\hat{A}$; the true error is the residual trace $\text{tr } R_A(J) = \|F - P_{F_J} F\|_F^2$ of the selected columns. In this section $\varepsilon \in (0, 1]$ denotes the target accuracy of RP; the distortion of the sketch is written $2\varepsilon_0$.

The frame of interest is chosen by the sketch, so a fixed-frame OSE does not apply directly, and the gate of Section 6 does not fit either: at full rank it is trivial, and at partial rank it is outside its hypotheses. The result below shows that RP needs embeddings only of the $(k + 1)$ -dimensional frames that it actually meets along its pivot path, and that these frames can be handled through the exact likelihood of the path.

Lemma 7.1 (Moment bound at every order). *Let (C, B) be one of the pairs of Theorem 3.1, let $q \geq 2$ and $0 < \varepsilon \leq 1$, and let $s = \lceil B(2q + 1)/\varepsilon \rceil$, $b = \lceil C(d + 2q + 1)/(s\varepsilon^2) \rceil$. Then every isometry U with $d' \leq d$ columns satisfies $\mathbb{E} \text{tr}(U^\top \Pi^\top \Pi U - I_{d'})^{2q} < d'(\varepsilon/2)^{2q}$.*

Proof. The case $d' = d$ is the case $q \geq 2$ of the proof of Theorem 3.1, which does not use the relation between q and δ . For $d' < d$, the numbers α_ν, χ_ν of (2)–(3) increase with d , so T_q for d' is entrywise at most T_q for d , and Imported Theorem 2.3 gives $\mathbb{E} \text{tr } E^{2q} \leq d'(T_q^{2q})_{00} < d'(\varepsilon/2)^{2q}$. $\square$

The following consequence uses the fixed-state comparison, hybrid argument and determinant route of *RPCholesky under approximate and sketched pivot probabilities* [8]. We state the basic trace recurrence, specialize its moment input to SparseStack, and record explicitly the moment order required by that determinant route.

Imported theorem 7.2 (Sketched RPCholesky [8]). *Let $0 < \eta \leq 1/4$, put $\gamma = (1 - \eta - \eta^2)/(1 + \eta)$, let $0 < \varepsilon_0 \leq \eta/2$ and $q \geq \max\{2, k\}$, and let Π be SparseStack with the prescription of Lemma 7.1 at dimension $d = k + 1$, with $2\varepsilon_0$ in place of ε . Let $\varphi_\gamma(x) = x - \gamma(x - \tau)_+^2/(rx)$. Then:*

- (i) $\mathbb{E} \operatorname{tr} R_A(J_k) \leq \varphi_\gamma^{\circ k}(\operatorname{tr} A) + \operatorname{tr} A \cdot P_{\text{fail}}$;
- (ii) $P_{\text{fail}} \leq \sum_{t=1}^k (1 - \eta - \eta^2)^{-t} \sqrt{t} (1 + \varepsilon_0)^t \sqrt{\delta_1}$, where $\delta_1 \leq (3k + 4)2^{-q}$ bounds the failure probability of the good event below for any fixed pivot set of size at most k ;
- (iii) on the good event at J_k , $\|F - F_J \hat{X}\|_F^2 \leq (1 + \eta^2/(2(1 - \eta)^2)) \operatorname{tr} R_A(J_k)$.

In particular, for $0 < \varepsilon \leq 1$, the choices $k \geq (r/\gamma)(1/\varepsilon + \log(1/\varepsilon) + \log(1/\eta_A))$ and $q = O(k + \log(1/(\varepsilon\eta_A)))$, with $\varepsilon_0 = \eta/2$, give $\mathbb{E} \operatorname{tr} R_A(J_k) \leq (1 + 2\varepsilon)\tau$, with

$$m = O\left(\frac{k + \log(1/(\varepsilon\eta_A))}{\eta^2}\right)$$

rows, independent of n and of $\operatorname{rank} F$. For $k < n$, if the sketched residual trace reaches zero before k pivots, continue with an arbitrary unselected index until the budget is used; the theorem concerns this completed rule. A premature zero state is a failure unless the true residual is also zero. If $k \geq n$, selecting all original columns and using the exact refit gives zero error.

Proof outline: establish one fixed-state event. For a fixed set J let Q_J be an orthonormal basis of range F_J , let $E_J = (I - Q_J Q_J^\top)F$ be the true residual, and let W_J be the diagonal matrix of drift weights $\|R_A e_i\|^2/(R_A)_{ii}$, taking the weight to be zero when the denominator is zero. For each nonzero column m_i of $M = E_J$ or $M = E_J W_J^{1/2}$, set $u_i = m_i/\|m_i\|$ and $Z_i = [Q_J, u_i]^\top \Pi^\top \Pi [Q_J, u_i] - I$. The good event $G(J)$ requires the embedding of Q_J with error at most η and, for each of these two matrices, the weighted bound

$$\Phi(M) = \sum_{i:m_i \neq 0} \frac{\|m_i\|^2}{\|M\|_F^2} \|Z_i\|^4 \leq \eta^4/4.$$

For $M = 0$ this condition is automatic. Jensen's inequality implies that $\|\Pi M\|_F^2$ lies within $(1 \pm \eta)\|M\|_F^2$ and $\|Q_J^\top \Pi^\top \Pi M\|_F^2 \leq (\eta^2/2)\|M\|_F^2$. Minkowski's inequality in $L^{q/2}$ and Markov's inequality give

$$\mathbb{P}\{G(J)^c\} \leq (k + 1)(4^{-q} + 2 \cdot 2^{-q}) \leq (3k + 4)2^{-q} =: \delta_1.$$

Each frame has dimension at most $k + 1$; the weighted averages avoid a union bound over columns.

Turn that event into true-residual progress. On $G(J)$ the identity $\operatorname{tr} R_{\hat{A}}(J) = \|\Pi E_J\|_F^2 - \|P_{\Pi F_J} \Pi E_J\|_F^2$ gives $(1 - \eta - \eta^2) \operatorname{tr} R_A \leq \operatorname{tr} R_{\hat{A}} \leq (1 + \eta) \operatorname{tr} R_A$, and the same identity applied to $E_J W_J^{1/2}$ shows that the sketched step decreases the true residual trace by at least $\gamma \operatorname{tr}(R_A^2)/\operatorname{tr} R_A$ in expectation.

Control the first failed state along the path. A hybrid process that follows sketched RP until the first bad state and exact RP afterwards obeys the γ -recurrence, and Jensen’s inequality for the concave map φ_γ gives (i). For (ii), the exact path identity

$$\frac{\text{pr}_\Pi(w)}{\text{pr}(w)} = \det(Q_J^\top \Pi^\top \Pi Q_J) \prod_{j < t} \frac{\text{tr } R_A(J_j)}{\text{tr } R_{\hat{A}}(J_j)},$$

for an ordered pivot path w of length t with prefix sets J_j and final set J , compares the sketched path law pr_Π with the exact RP path law pr , which does not depend on Π ; it holds because the product of pivot diagonals equals a determinant. Cauchy–Schwarz with $\mathbb{E} \det(Q_J^\top \Pi^\top \Pi Q_J)^2 \leq t(1 + \varepsilon_0)^{2t}$ then bounds the probability that the first failure occurs at step t . This uses the $2q$ -th Gram-error moment to control a determinant of degree t , and requires $t \leq q$. Lemma 7.1 supplies the actual frame-dimension prefactor t , giving the displayed $\sqrt{t}$ factor. The stronger hybrid likelihood bound in [8] dispenses with the restriction $q \geq k$.

Recover the same-sketch regression error. Write $F_J = QT$ with $Q = Q_J$ and T of full row rank, and let $E = E_J$. On the good event, $G = Q^\top \Pi^\top \Pi Q$ is invertible. The prediction, rather than the coefficient matrix, satisfies

$$F_J \hat{X} = QQ^\top F + QG^{-1}Q^\top \Pi^\top \Pi E.$$

The two residual components are orthogonal because $Q^\top E = 0$. Consequently the excess squared error is $\|G^{-1}Q^\top \Pi^\top \Pi E\|_F^2$, which yields (iii).

Remark 7.3. The row count must grow with the pivot budget k , not only with r : RP on $\hat{A}$ stops after rank $\hat{A} \leq m$ pivots. Numerically, sketched RP with $m = 2k$ SparseStack rows already matches exact RP closely (Section 8.5); the provable constants are far more conservative because of the $e^{O(k)}$ likelihood factor in (ii).

8 Numerical diagnostics

The diagnostics in this section are numerical evidence, not proofs. The original scripts and recorded outputs are supplied in this paper’s `checks/` directory. They were run in Julia 1.13; majorant computations use 128–256-bit floating point, Nyström and diagonal experiments use 2×10^4 – 4×10^4 trials, gate experiments enumerate all pivot paths, and the final sketched-RP experiment uses 150 trials per setting. Only the exact rational certificate of Proposition 3.5 enters a proof.

An independent Python check, supplied as `checks/independent_audit.py`, reproduces all 63 rational certificates and the integer parameter table, enumerates 64 small CountSketch matrices to verify the moment and drift calculations, and enumerates full-rank RP laws on 48 complex instances. Its separate log-domain majorant propagation reproduces the displayed maxima 0.352 and 0.477 for $C = 69$ and $C = 35$. These finite numerical checks supplement the analytic arguments; they do not establish the open all-order conjecture.

8.1 Headroom in the scalar majorant

With the prescription of Theorem 3.1 and varying C at $B = 8$, we built T_q from (2)–(3) and iterated it in high precision, computing $\rho = (T_q^{2q})_{00}^{1/(2q)} / \varepsilon$ over $d \in \{1, 10, 10^2, 10^4, 10^6, 10^8\}$,

$q \in \{2, 5, 10, 40, 100, 200\}$ and $\varepsilon \in \{1, 1/2, 1/20\}$; for $C \in \{69, 35\}$ the grid was extended to $d \in \{10, 10^3, 10^6, 10^9\}$ and $q \in \{200, 1000, 3000\}$ at $\varepsilon = 1$. The proof needs $\rho < 1/2$. The worst values occur at $\varepsilon = 1$ and large d :

(C, B)	(81, 17/2)	(69, 8)	(40, 8)	(35, 8)	(30, 8)
$\max \rho$	0.319	0.352	0.441	0.477	0.505

At $(C, B) = (69, 8)$ the envelope certificate is tight ($T_{\ell, r}/10^6 \leq 0.4988$), while the majorant itself sits near 0.35. The threshold $1/2$ is crossed between $C = 30$ and $C = 35$, which motivates Conjecture 3.7. On a grid of $d \leq 10^6$, $q \leq 40$ and five values of ε , the row sums of T_q/ε never exceeded $\Phi_{t, b}(u_k)$, in agreement with Lemma 3.3.

8.2 Nyström and drift experiments

Table 3 compares, for SparseStack, the Monte Carlo mean of $\text{tr}(AZAZ)$ with formula (12), the realized $\mathbb{E}\|R\|_F^2$, and the two upper bounds $2(\text{tr } A)^2/m$ (Theorem 4.2) and $\kappa_b(\text{tr } A)^2/s$ (Corollary 5.4). The deterministic inequality $\|R\|_F^2 \leq \text{tr}(AZAZ)$ held in every trial. The lower part of the table gives the one-block drift experiments of Remark 5.5.

Matrix, (s, b)	MC $\text{tr}(AZAZ)$	formula (12)	$\mathbb{E}\ R\ _F^2$	$2(\text{tr } A)^2/m$	$\kappa_b(\text{tr } A)^2/s$
$A_{30}, (4, 6)$	0.568	0.568	$< 10^{-4}$	1.019	4.077
Wishart 12×12 , $(3, 4)$	8.103	8.059	0.043	16.445	49.334
$I_{20}, (2, 5)$	37.988	38.000	10.116	80.000	280.000
One block, drift	observed	bound	ratio		
$R = I_{12}, b = 3$	2.978	0.600	4.96		
Wishart 12×12 , $b = 3$	4.272	0.993	4.30		
$R = I_{40}, b = 8$	7.963	0.800	9.95		
rank one, $n = 12$, $b = 3$	1.000	0.600	1.67		

Table 3: Monte Carlo diagnostics (2×10^4 trials) for Sections 4 and 5, with $A_{30} = G \text{diag}(0.7^j) G^\top / 30$. “Bound” is $\text{tr}(R^2)/(\kappa_b \text{tr } R)$ from Theorem 5.3. For I_{20} with $m = 10$, $\|R\|_F^2 = 20 - \text{rank } \Pi \geq 10$, so the rank-blind bound cannot be small there.

The diagonal-training formula of Theorem 5.7 was checked on a 12×12 Wishart matrix with $(s_0, b) = (3, 4)$: Monte Carlo mean squared error 0.6045 against the exact value 0.6036 (4×10^4 trials).

8.3 The gate

Exhaustive enumeration of the full-rank RP set law on VCV^* for 40 random complex instances ($N \in \{4, 5, 6\}$, $k \in \{2, 3\}$, random isometry V and random positive-definite C) gave $\max_J \mu_C(J)/(\Lambda_C \nu_V(J)) = 0.961$, consistent with Proposition 6.1; equality $\mu_C = \nu_V$ holds for flat C . The spike formula of Lemma 6.2 was confirmed at $k = 10$, $\eta = 3$: $\Lambda_C = 40/13 \approx 3.077$.

8.4 Factorial versus Rényi prices

Figure 4 plots the ratio of the two failure-allowance costs, $[3 \log(1/\delta) + \mathfrak{d}_k]/[\log(1/\delta) + \log k!]$ at $\delta = 0.01$, using the optimal α in $B_{k,2/3}$. The factorial gate is cheaper as long as the ratio exceeds one.

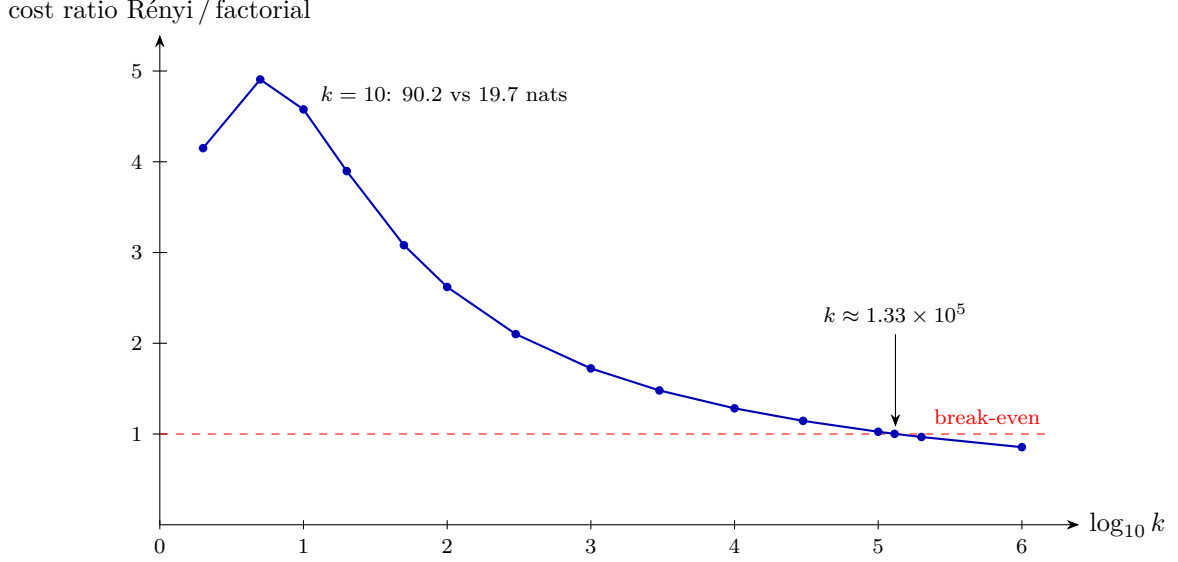


Figure 4: Ratio of the Rényi-order-3/2 cost to the factorial cost of the failure allowance, $\delta = 0.01$ (numerical evaluation of exact formulas). Below $k \approx 1.3 \times 10^5$ the elementary $k!$ price is cheaper.

8.5 Sketched RPCholesky

We took $F = U\Sigma V^\top \in \mathbb{R}^{300 \times 400}$ with Haar factors, five unit singular values and 295 squared singular values in $[2.5, 5] \times 10^{-6}$, so $r = 5$ and $\eta_A \approx 2.2 \times 10^{-4}$, and compared exact RP on $A = F^\top F$ with RP on $(\Pi F)^\top (\Pi F)$ for a SparseStack with $s = 4$ blocks and for a Gaussian sketch, 150 trials each. The table reports $\mathbb{E} \operatorname{tr} R_A(J_k)/\tau$ and the mean ratio of the same-sketch regression error $\|F - F_J \hat{X}\|_F^2$ to $\operatorname{tr} R_A(J_k)$.

	exact RP	SparseStack RP	Gaussian RP	regression ratio	$1 + \frac{k}{m-k-1}$
$k = 10, m = 20$	1.610	1.632	1.624	1.98	2.11
$k = 10, m = 40$	1.610	1.664	1.645	1.34	1.35
$k = 20, m = 40$	1.195	1.201	1.199	1.98	2.05
$k = 20, m = 80$	1.195	1.196	1.199	1.33	1.34

Already at $m = 2k$ the sketched pivots are within a few percent of exact ones, whereas the same-sketch regression pays roughly the Gaussian factor $1 + k/(m - k - 1)$. Selecting pivots is therefore much less demanding than solving the sketched regression, which is consistent with the path-local structure of Imported Theorem 7.2.

9 Open problems

1. *The scalar step.* Prove Conjecture 3.7, or any bound on returning walks of T_q sharper than the row-sum product of Lemma 2.4. The diagnostics of Section 8.1 suggest that the operator estimates of the imported theorems already support a row constant near 86 in the form $(2C + 2B)$, and a d -coefficient near 35. A certificate would likely combine a diagonal similarity of T_q with an exact interval partition in $t = v/D$, as in Section 3.2.
2. *Block progress at the right scale.* Replace $\text{tr } F$ by $\lambda_{\max}(F)$ in the proof of Theorem 5.3, through a block-level size-biased ratio controlled by a mixed moment of the top eigenvalue of $HRH^\top$, so that a b -column block is credited with progress proportional to $\min(b, \text{rank } R)$ probes.
3. *Tail-relative high-probability Nyström bounds.* Proposition 4.4 excludes uniform expected tail-relative guarantees for finite-support sketches with $m < n$. Determine the sharp high-probability dependence of $\|A - \mathcal{N}_A(\Pi)\|$ on the tail for SparseStack, in terms of m , s and δ , without passing through an OSE for the leading eigenspace.
4. *A useful selection rule with a fixed range.* Proposition 6.1 requires the frame V and the dictionary to be fixed in advance and the selection to be full rank. Find natural selection rules in which a sketch-dependent $C(S)$ carries real information while V stays fixed, or extend the likelihood argument to partial-rank stopping.
5. *Gate prices between $k!$ and Rényi.* For $10 \leq k \leq 10^5$, neither the factorial nor the Rényi price is sharp. A pointwise likelihood bound that interpolates between $k!$ and $\exp(O(k \log \log \log k))$, possibly using the spectrum of $C(S)$ as in Lemma 6.2, would lower q for moderate selector ranks.
6. *Sketched RP constants.* The likelihood factor $e^{O(k)}$ in Imported Theorem 7.2(ii) makes the provable row count about $10^3\text{--}10^4 \cdot k$ at constant η , while Section 8.5 shows $m = 2k$ working in practice. Remove the exponential factor, for instance by a martingale argument along the pivot path.

A The certificate checker

The following self-contained Julia program (standard library only) reproduces Proposition 3.5 and Table 1. All arithmetic is exact: `Rational{BigInt}` throughout, each square root is replaced by an upper bound $\lceil 10^{12} \sqrt{x} \rceil / 10^{12}$ obtained from an integer square root and verified by squaring, and the final comparisons are integer comparisons. An integer bisection finds $T_{\ell,r}$, so no floating-point operation enters the certificate. The program prints one line per pair and interval and ends with `certificate valid: true`.

In words, for each of the 3×21 cases the program: (1) computes $n_{\ell,r}$ from (9); (2) evaluates the upper-rounded envelope (10) exactly at the eight nodes $u_j = (2j + 1)/16$ and multiplies the values into $P_{\ell,r}$; (3) checks $P_{\ell,r} < 1/256$; (4) finds the least integer T with $T^8 \geq 10^{48} P_{\ell,r}$ and checks $T < 500000$; and (5) checks the vacuum condition $9C < 800$.

```
# Exact certificate for the interval envelopes Psi_{l,r}: every radical is
# rounded up to a rational with denominator K; everything else is exact.
const K = big(10)^12
function sqrt_up(x::Rational{BigInt})          # a rational >= sqrt(x)
```

```

    N, D = numerator(x) * K^2, denominator(x)
    k = isqrt(cld(N, D))
    while k * k * D < N; k += 1; end
    return k // K
end
const S2 = sqrt_up(big(2) // 1)
G_up(n) = (2 * sqrt_up((n - 1) // n) + 1) * sqrt_up(1 // n)
function Psi_up(u, l, r, C, B, n)
    th = 1 - l + l * u
    2 * sqrt_up((1 + r * u / C) * th / C) + (2 + S2) * sqrt_up(u * th / (C * B)) +
    th / C + (1 + S2 + G_up(n)) * u / B
end
bucket(C, B, r) = max(floor(BigInt, C / B), floor(BigInt, C / (r * (B + 1 // 5)))) + 1
intervals = [[(big(j) // 40, big(j + 1) // 40) for j in 0:19]; (big(1) // 2, big(1) // 1)]
pairs = [(big(69) // 1, big(8) // 1), (big(74) // 1, big(15) // 2), (big(79) // 1, big(7) //
1)]
nodes = [big(2j + 1) // 16 for j in 0:7]
ok = true
for (C, B) in pairs, (l, r) in intervals
    n = bucket(C, B, r)
    P = prod(Psi_up(u, l, r, C, B, n) for u in nodes)    # >= prod of Psi(u_j)
    lo, hi = big(0), big(500000)
    while hi - lo > 1
        mid = (lo + hi) >> 1
        if mid^8 >= P * big(10)^48; hi = mid; else; lo = mid; end
    end
    T = hi
    # least T with (T/1e6)^8 >= P, if P < 1/256
    global ok &= (P < big(1) // 256) && (T < 500000) && (9C < 800)
    println("C=$C B=$B [$l,$r] n=$n T=$T")
end
println("certificate valid: ", ok)

```

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